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Capacitive touch panel and display device

Abstract

A touch panel includes: a drive electrode supplied with a drive signal; a position detecting electrode detecting a position included in a touch surface and touched with a pointing object, and a pressure detecting electrode detecting pressure applied on the touch surface with the pointing object, the position detecting electrode and the pressure detecting electrode being disposed across the drive electrode from the touch surface; and a floating electrode having a floating potential. The position detecting electrode, the pressure detecting electrode, and the floating electrode are formed in a layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of priority to Japanese Patent Application Number 2022-115046 filed on Jul. 19, 2022. The entire contents of the above-identified application are hereby incorporated by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a capacitive touch panel and a display device.

BACKGROUND ART

(3) A conventionally known capacitive touch panel includes a pressure detecting electrode and a position detecting electrode. Such a capacitive touch panel is disclosed in, for example, Japanese Unexamined Patent Application Publication No. 2021-128511.

(4) The capacitive touch panel disclosed in Japanese Unexamined Patent Application Publication No. 2021-128511 includes: a drive electrode formed on a first substrate; a position sensing

electrode formed on a second substrate; and a pressure sensing electrode. On this capacitive touch panel, a pointing object is capacitively coupled to the drive electrode and the position sensing electrode. As a result, an electrostatic capacitance between the drive electrode and the position sensing electrode decreases, and a signal from the position sensing electrode changes. In accordance with the change of the signal from the position sensing electrode, a position of the pointing object is detected. Moreover, when the capacitive touch panel is pressed with the pointing object, and a distance between the drive electrode and the pressure sensing electrode decreases, the electrostatic capacitance between the drive electrode and the pressure sensing electrode increases. Accordingly, a signal from the pressure sensing electrode changes. Thus, in accordance with the change of the signal from this pressure sensing electrode, a magnitude of the pressure is detected.

SUMMARY

Technical Problem

(5) Here, in the capacitive touch panel including the position detecting function and the pressure detecting function described in Japanese Unexamined Patent Application Publication No. 2021-128511, an electrostatic capacitance is formed between the drive electrode and the position sensing electrode, and additionally between the drive electrode and the pressure sensing electrode.

Moreover, the position sensing electrode and the pressure sensing electrode are formed in the same layer (hereinafter referred to as a “layer in which a detection electrode is formed”). As a result, compared with a capacitive touch panel having no pressure detecting function, the capacitive touch panel of Japanese Unexamined Patent Application Publication No. 2021-128511 exhibits an increase in the electrostatic capacitance between the drive electrode and an electrode that is formed in the layer in which a detection electrode is formed. This inevitably results in an increase in a time constant of a drive signal to be supplied to the drive electrode, which might take a long response time to detect the touch and the pressure (might cause poor response characteristics).

(6) Therefore, the present disclosure is intended to solve the above problems and to provide a capacitive touch panel and a display device having a pressure detecting function capable of preventing an increase in an electrostatic capacitance between a drive electrode and an electrode that is formed in a layer in which a detection electrode is formed.

Solution to Problems

(7) In order to solve the above problems, a capacitive touch panel according to a first aspect of the present disclosure detects: a position included in a touch surface and touched with a pointing object; and pressure applied on the touch surface with the pointing object. The capacitive touch panel includes: a drive electrode supplied with a drive signal; a detection electrode disposed across the drive electrode from the touch surface, and including a position detecting electrode that detects the position included in the touch surface and touched with the pointing object, and a pressure detecting electrode that detects the pressure applied on the touch surface with the pointing object; and a floating electrode having a floating potential. The position detecting electrode, the pressure detecting electrode, and the floating electrode are formed in a layer.

(8) Moreover, a display device according to a second aspect includes: a capacitive touch panel that detects: a position included in a touch surface and touched with a pointing object; and pressure applied on the touch surface with the pointing object; and a display that displays an image. The capacitive touch panel includes: a drive electrode supplied with a drive signal; a detection electrode disposed across the drive electrode from the touch surface, and including a position detecting electrode that detects the position included in the touch surface and touched with the pointing object, and a pressure detecting electrode that detects the pressure applied on the touch surface with the pointing object; and a floating electrode having a floating potential. The position detecting electrode, the pressure detecting electrode, and the floating electrode are formed in a layer.

Advantageous Effects

(9) According to the above configuration, the floating electrode having a floating potential is formed in the same layer as a layer in which a detection electrode is formed. Between the drive

electrode and the floating electrode, an electrostatic capacity is less likely to be formed. Hence, even if the capacitive touch panel has a pressure detecting function, such a feature makes it possible to prevent the electrostatic capacity from increasing between the drive electrode and an electrode that is formed in the layer in which a detection electrode is formed.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a cross-sectional view of a schematic configuration of a display device **100** according to a first embodiment.
- (2) FIG. 2 is a schematic plan view of a configuration of the display device **100** according to the first embodiment.
- (3) FIG. 3 is a cross-sectional view of a touch panel **1** according to the first embodiment.
- (4) FIG. 4 is a plan view of a portion of a drive electrode layer **11**.
- (5) FIG. 5 is a plan view of a portion of a detection electrode layer **21**.
- (6) FIG. 6 is a view of a configuration of a display device **200** according to a first comparative example.
- (7) FIG. 7 is a table showing a result of comparison between the display device **100** according to an example of the first embodiment and the display device **200** according to the first comparative example.
- (8) FIG. 8 is a graph showing a result of comparison between the intensity of a signal from a pressure detecting electrode **21a** according to an example of the first embodiment and the intensity of a signal from a pressure detecting electrode according to the first comparative example.
- (9) FIG. 9 is a cross-sectional view of a touch panel **301** included in a display device **300** according to a second embodiment.
- (10) FIG. 10 is a plan view of a portion of a detection electrode layer **321** according to the second embodiment.
- (11) FIG. 11 is a table showing a result of comparison between the display device **300** according to an example of the second embodiment and the display device **200** according to the first comparative example.
- (12) FIG. 12 is a plan view of a portion of a display device **400** according to a third embodiment.
- (13) FIG. 13 is a view of a configuration of a display device **500** according to a second comparative example.
- (14) FIG. 14 is a table showing a result of comparison between the display device **400** according to an example of the third embodiment and the display device **500** according to the second comparative example.

DESCRIPTION OF EMBODIMENTS

(15) Embodiments of the present disclosure will be described below with reference to the drawings. Note that the present disclosure shall not be limited to the embodiments below, and the designs presented in the embodiments can be appropriately modified within a scope of the configurations of the present disclosure. Moreover, identical reference signs are used to denote identical or substantially identical components among the drawings. Such components will not be repeatedly elaborated upon. Furthermore, the configurations described in the embodiments and in a modification thereof may be appropriately combined or modified unless otherwise departing from the scope of the present disclosure. For the sake of convenience, the drawings below are simplistically or schematically illustrated. In the drawings, some of the constituent members may be omitted. In addition, the dimensional ratios between the constituent members in the drawings are not necessarily the actual dimensional ratios.

First Embodiment

(16) (Configuration of Display Device)

(17) FIG. 1 is a cross-sectional view of a schematic configuration of a display device **100** according to a first embodiment. The display device **100** includes: a touch panel **1**; and a display unit **2** that displays an image on a display surface **2a**. The display unit **2** is, for example, a liquid crystal display or an organic electroluminescence (EL) display.

(18) (Configuration of Touch Panel)

(19) FIG. 2 is a schematic plan view of a configuration of the display device **100** according to the first embodiment. The display device **100** includes a touch panel **1** that detects a position touched with a pointing object, a position pressed with the pointing object, and a magnitude of the pressure. As illustrated in FIG. 2, the touch panel **1** includes: a plurality of drive electrodes **11a**; a plurality of pressure detecting electrodes **21a**; and a plurality of position detecting electrodes **21b**. The touch panel **1** is a mutual capacitive touch panel that detects a change in capacitance between the plurality of drive electrodes **11a** and either the plurality of pressure detecting electrodes **21a** or the plurality of position detecting electrodes **21b**. Moreover, the display device **100** includes a touch panel controller **3**. The touch panel controller **3** is connected through wires **3a** to each of the plurality of drive electrodes **11a**, the plurality of pressure detecting electrodes **21a**, and the plurality of position detecting electrodes **21b**. The touch panel controller **3** is, for example, an integrated circuit.

(20) FIG. 3 is a cross-sectional view of the touch panel **1** according to the first embodiment. As illustrated in FIG. 3, the touch panel **1** includes: a first substrate **10**; a pressure-sensitive layer **4**; and a second substrate **20**. The first substrate **10**, the pressure-sensitive layer **4**, and the second substrate **20** are stacked on top of another. For example, the first substrate **10** and the second substrate **20** are formed of a transparent resin material such as polyethylene terephthalate (PET). Moreover, the pressure-sensitive layer **4** is a dielectric film containing a dielectric material. Furthermore, the pressure-sensitive layer **4** is made of an elastic transparent pressure-sensitive material such as a polymer material. Then, the touch panel **1** includes a cover member **5**. The cover member **5** is made of, for example, a glass material. Moreover, an optical clear adhesive (OCA) layer **12** is disposed between the first substrate **10** and the cover member **5**. A surface (a touch surface) of the cover member **5** is touched and pressed with the pointing object. The OCA layer **12** bonds the cover member **5** and the first substrate **10** together. Furthermore, an OCA layer **22** is disposed between the second substrate **20** and the display unit **2**.

(21) As illustrated in FIG. 3, the touch panel **1** includes: a drive electrode layer **11**; and a detection electrode layer **21**. The drive electrode layer **11** is formed on a surface **10a** of the first substrate **10** toward the pressure-sensitive layer **4**. The detection electrode layer **21** is formed on a surface **20a** of the second substrate **20** toward the pressure-sensitive layer **4**. That is, the detection electrode layer **21** is disposed across the drive electrode layer **11** from the touch surface. The drive electrode layer **11** includes: the drive electrodes **11a**; and drive floating electrodes **11b**. The detection electrode layer **21** includes: the position detecting electrodes **21b**; the pressure detecting electrodes **21a**; reference potential electrodes **21c**; and floating electrodes **21d**. Moreover, the drive electrodes **11a**, the drive floating electrodes **11b**, the position detecting electrodes **21b**, the pressure detecting electrodes **21a**, the reference potential electrodes **21c**, and the floating electrodes **21d** are made of a transparent conductive material such as indium tin oxide (ITO). Furthermore, the reference potential electrodes **21c** are connected to a not-shown ground, and the potential of the reference potential electrodes **21c** is equal to the potential of the ground. Each of the drive floating electrodes **11b** and the floating electrodes **21d** is not connected to another electrode, not connected to a voltage source (e.g., the touch panel controller **3**), or not connected to the ground. Each of the drive floating electrodes **11b** and the floating electrodes **21d** has a floating potential. The floating potential is a potential that varies depending on the surrounding electric field because the floating potential is not connected to another conductive member.

(22) As illustrated in FIG. 3, a pressure detecting electrode **21a** is positioned to overlap with a drive electrode **11a** in plan view. Moreover, at least a portion of a position detecting electrode **21b** is

positioned not to overlap with the drive electrodes **11a** in plan view. A floating electrode **21d** is positioned to overlap with the drive electrode **11a** in plan view. Furthermore, the position detecting electrode **21b** is positioned to overlap with a drive floating electrode **11b** in plan view. In addition, the floating electrode **21d** is positioned not to overlap with the drive floating electrode **11b** in plan view. Moreover, in the first embodiment, a reference potential electrode **21c** is disposed between the floating electrode **21d** and the position detecting electrode **21b**.

(23) The drive electrode **11a** is supplied with a drive signal from the touch panel controller **3** (see FIG. 2). Moreover, as illustrated in FIG. 3, the drive floating electrode **11b** forms a capacitance C_a (an electrostatic capacitance of a fringe component) between the drive floating electrode **11b** and the drive electrode **11a**. The position detecting electrode **21b** is an electrode for detecting a position touched with the pointing object. The position detecting electrode **21b** forms a capacitance C_b between the position detecting electrode **21b** and the drive floating electrode **11b**. The pressure detecting electrode **21a** is an electrode for detecting presence or absence of, and a magnitude of, the pressure applied with the pointing object. The pressure detecting electrode **21a** forms a capacitance C_c between the pressure detecting electrode **21a** and the drive electrode **11a**. The reference potential electrode **21c** has a ground potential of the touch panel **1**. The reference potential electrode **21c** forms a capacitance C_d between the reference potential electrode **21c** and the drive electrode **11a**. Here, if the drive floating electrode and the pressure detecting electrode are capacitively coupled together, the capacitance inevitably changes just because the pointing object touches the touch panel even if the touch panel is not pressed by the pointing object. In this case, it is difficult to distinguish the capacitance change caused merely by a touch of the pointing object from the capacitance change caused by a press of the pointing object. On the other hand, according to the configuration of the first embodiment, the reference potential electrode **21c** can prevent capacitive coupling between the drive floating electrode **11b** and the pressure detecting electrode **21a**. Hence, the drive electrode **11a** and the pressure detecting electrode **21a** exhibit the capacitance change only when the touch panel **1** is pressed. As a result, according to the first embodiment, it is possible to accurately distinguish the capacitance change caused merely by a touch of the pointing object from the capacitance change caused by a press of the pointing object. The floating electrode **21d** forms a capacitance C_e (an electrostatic capacitance of a fringe component) between the floating electrode **21d** and the pressure detecting electrode **21a**.

(24) Thanks to the above features, an electrostatic capacitance is less likely to be formed between the drive electrode **11a** and the floating electrode **21d**. Compared with a case where the entire detection electrode layer is composed only of position detecting electrodes, pressure detecting electrodes, and the reference potential electrodes, the features can reduce an electrostatic capacitance between the drive electrode **11a** and an electrode formed in the detection electrode layer **21**. As a result, the touch panel **1** having a pressure detecting function can prevent an increase in the electrostatic capacitance between the drive electrode **11a** and the electrode formed in the detection electrode layer **21**. This successfully prevents an increase in a time constant of a drive signal to be supplied to the drive electrode **11a**, and a delay of a response time to detect the touch and the pressure (poor response characteristics).

(25) Moreover, the pressure detecting electrode **21a**, the reference potential electrode **21c**, and the drive electrode **11a** constitute a parallel plate capacitor. The position detecting electrode **21b** and the drive electrode **11a** form an electrostatic capacitance (an electrostatic capacitance of a fringe component) between an end portion of the drive electrode **11a** and the position detecting electrode **21b**. An electrostatic capacitance formed by the parallel plate capacitor is larger than the electrostatic capacitance of the fringe component. On the other hand, thanks to the above configuration, at least a portion of an electrode, included in the detection electrode layer **21** and positioned to overlap with the drive electrode **11a** in plan view, acts as the floating electrode **21d**. Such a feature can reduce the size of the parallel plate capacitor. As a result, the feature can prevent an increase in the electrostatic capacitance between the drive electrode **11a** and an electrode formed

in the detection electrode layer **21**.

(26) Moreover, as described above, the drive electrode **11a** and the drive floating electrode **11b** are capacitively coupled, and the drive floating electrode **11b** and the position detecting electrode **21b** are capacitively coupled. Thanks to such a feature, even if the drive electrode **11a** and the position detecting electrode **21b** are positioned not to overlap with each other in plan view, the drive electrode **11a** and the position detecting electrode **21b** can form an electrostatic capacitance through the drive floating electrode **11b**. As a result, the feature can increase the intensity of a signal for detecting the touch position based on the electrostatic capacitance. Moreover, the floating electrode **21d** is positioned not to overlap with the drive floating electrode **11b** in plan view, such that the position detecting electrode **21b** can be formed in a large size and positioned to overlap with the floating electrode **11b** in plan view. These features make it possible to reduce a difference between the intensity of a signal from the pressure detecting electrode **21a** and the intensity of a signal from the position detecting electrode **21b**, so that the intensity of the signal from the pressure detecting electrode **21a** and the intensity of the signal from the position detecting electrode **21b** can be easily adjusted on the touch panel controller **3**.

(27) FIG. 4 is a plan view of a portion of the drive electrode layer **11**. As illustrated in FIG. 4, the drive electrode **11a** includes a plurality of electrode units **51** shaped into rectangles and connected together in a horizontal direction (in an X-direction) on the drawing. The two adjacent electrode units **51** are connected by, for example, two connecting portions **11c** extending in the X-direction. Each of the plurality of drive floating electrodes **11b** is not connected to another drive floating electrode **11b**. Each drive floating electrode **11b** is a rectangle electrode. The drive floating electrode **11b** is disposed in the X-direction between two adjacent electrode units **51**. Moreover, between two connection portions **11c** in the Y direction, two drive floating electrodes **11b** are arranged. A length **L22** of each drive floating electrode **11b** in the X-direction is shorter than a length **L12** of each electrode unit **51** in the X-direction. A length **L21** of each drive floating electrode **11b** in a Y-direction is shorter than a length **L11** of each electrode unit **51** in the Y-direction. Furthermore, as illustrated in FIGS. 2 and 3, the drive electrode **11a** is positioned to overlap with the pressure detecting electrode **21a**, the reference potential electrode **21c**, and the floating electrode **21d** in plan view. In addition, the drive floating electrode **11b** is positioned to overlap with the position detecting electrode **21b** in plan view. Moreover, as illustrated in FIG. 4, each of the plurality of electrode units **51** is larger in area than a drive floating electrode **11b**.

(28) FIG. 5 is a plan view of a portion of the detection electrode layer **21**. As illustrated in FIG. 5, in the detection electrode layer **21**, the pressure detecting electrodes **21a** and the position detecting electrodes **21b** are alternately arranged in the horizontal direction (in the X-direction) on the drawing. Each of the plurality of pressure detecting electrodes **21a** and the plurality of position detecting electrodes **21b** extends in the vertical direction (in the Y-direction) on the drawing. Moreover, between a pressure detecting electrode **21a** and a position detecting electrode **21b**, a reference potential electrode **21c** is formed.

(29) Furthermore, as illustrated in FIG. 5, each of the plurality of pressure detecting electrodes **21a** includes a plurality of pressure detecting units **61** arranged side by side in the Y-direction. Each of the plurality of the pressure detecting units **61** in the first embodiment is shaped into a substantially rhombic hexagon. The plurality of pressure detecting units **61** each independently detect pressure of the pointing object. Each of the plurality of pressure detecting units **61** corresponds to one of the pressure positions (pressure coordinates) to be detected by the touch panel controller **3**. In other words, each pressure detecting unit **61** represents one of the pressure coordinates (a unit cell). Two adjacent pressure detecting units **61** in the Y-direction are connected by a connection portion **61a**. In the X-direction, the pressure detecting unit **61** has a largest width; namely a width **W31**, and a smallest width; namely, a width **W32**. Moreover, the width of the connecting portion **61a** is equal to the smallest width of the pressure detecting unit **61**.

(30) Furthermore, each of the plurality of position detecting electrodes **21b** includes a plurality of

position detecting units **62** arranged side by side in the Y-direction. The plurality of position detecting units **62** each independently detect a position touched with the pointing object. Each of the plurality of position detecting units **62** corresponds to one of the touched positions (touch coordinates) to be detected by the touch panel controller **3**. In other words, each position detecting unit **62** represents one of the touched position coordinates (a unit cell). Each position detecting unit **62** is shaped into a rectangle in plan view. In the X-direction, the position detecting unit **62** has a width of **W5**. For example, the width **W5** is larger than the width **W32** and smaller than the width **W31**.

(31) Moreover, the plurality of reference potential electrodes **21c** extend in the Y-direction along the position detecting electrodes **21b**. The plurality of reference potential electrodes **21c** are arranged between the floating electrodes **21d** and the position detecting electrodes **21b** in plan view. Furthermore, the plurality of reference potential electrodes **21c** are arranged to sandwich the position detecting electrodes **21b** in plan view. For example, each of the reference potential electrodes **21c** has a constant width **W6**. The width **W6** is smaller than, for example, the width **W5**.

(32) Moreover, the plurality of floating electrodes **21d** are disposed between the pressure detecting electrodes **21a** and the reference potential electrodes **21c**. Each of the plurality of floating electrodes **21d** in the first embodiment is shaped into a trapezoid. Furthermore, the plurality of floating electrodes **21d** are arranged side by side in the Y-direction. In addition, each of the plurality of floating electrodes **21d** is not connected to another floating electrode **21d**. Moreover, in the X-direction, each floating electrode **21d** has a largest width; namely, a width **W41**, and a smallest width; namely, a width **W42**. For example, the width **W41** is larger than the width **W6** and smaller than the width **W31**. In addition, the width **W42** is smaller than the width **W5** and larger than the width **W32**.

(33) (Operation of Touch Panel 1)

(34) Next, an operation of the touch panel **1** will be described, with reference to FIG. 3. As illustrated in FIG. 3, a drive floating electrode **11b** forms the capacitance **Cb** between the drive floating electrode **11b** and a position detecting electrode **21b**. When a pointing object **F** comes into contact with the cover member **5**, the pointing object **F** is capacitively coupled to each of the drive floating electrode **11b** and a drive electrode **11a**. As a result, the electrostatic capacitances (**Ca** and **Cb**) between the drive electrode **11a** and the position detecting electrode **21b** decrease through the pointing object **F** and the drive floating electrode **11b**, and the signal detected by the position detecting electrode **21b** (a position detecting unit **62**) changes. The touch panel controller **3** detects a position of the pointing object **F** in accordance with the change of the signal. Note that, at this time, the drive electrode **11a** is positioned between the pointing object **F** and a pressure detecting electrode **21a**. Hence, the drive electrode **11a** functions as a shielding material, and the capacitance **Cc** between the drive electrode **11a** and the pressure detecting electrode **21a** (the pressure detecting unit **61**) hardly changes.

(35) Moreover, as illustrated in FIG. 3, when the pointing object **F** presses the cover member the pressure-sensitive layer **4** is depressed, and a distance between the drive electrode **11a** and the pressure detecting electrode **21a** (the pressure detecting unit **61**) decreases. As a result, the capacitance **Cc** between the drive electrode **11a** and the pressure detecting electrode **21a** increases, and a signal detected by the pressure detecting electrode **21a** (the pressure detecting unit **61**) changes. Then, in accordance with the change of the signal (a pressure signal) to be detected by the pressure detecting electrode **21a**, the touch panel controller **3** detects a magnitude of the pressure (a pressure value). Note that, when the touch panel **1** is pressed, the distance between the drive floating electrode **11b** and the position detecting electrode **21b** decreases, and the capacitance **Cb** increases. At this time, a signal (a touch signal) from the position detecting electrode **21b** increases. However, the touch signal is sufficiently larger than the pressure signal, and is hardly affected. Thus, when the drive electrode **11a** is pressed with the pointing object **F**, the touch position can be detected highly precisely. Moreover, the detection electrode layer **21** is provided with the floating

electrode **21d**. Such a feature makes it possible to prevent an increase in the electrostatic capacitance C_d formed by the reference potential electrode **21c** and the drive electrode **11a**.
(36) Next, with reference to FIGS. **5** to **8**, a result of comparison is described between the display device **100** according to an example of the first embodiment and a display device **200** according to a first comparative example. Note that the display device **200** according to the first comparative example is configured to confirm advantageous effects of the display device **100** according to the example of the first embodiment. The display device **200** does not represent the prior art. FIG. **6** is a view of a configuration of the display device **200** according to the first comparative example.
(37) The example of the first embodiment is the display device **100** including the detection electrode layer **21** illustrated in FIG. **5**. Moreover, as illustrated in FIG. **6**, the display device **200** includes: a pressure detecting electrode **221a**; a position detecting electrode **221b**; and a reference potential electrode **221c**. In the first comparative example, the floating electrode **21d** is replaced with the reference potential electrode **221c**. The pressure detecting electrode **221a** and the position detecting electrode **221b** are respectively the same as the pressure detecting electrode **221a** and the position detecting electrode **221b**.

(38) FIG. **7** is a table showing a result of comparison between the display device **100** according to the example of the first embodiment and the display device **200** according to the first comparative example. As to each of the display device **100** according to the example of the first embodiment and the display device **200** according to the first comparative example, a value of the electrostatic capacitance C_b between the drive electrode and the position detecting electrode, a value of the electrostatic capacitance C_c between the drive electrode and the pressure detecting electrode, and a value of the electrostatic capacitance C_d between the drive electrode and the reference potential electrode were measured. The electrostatic capacitance C_b was 0.8 pF in the example of the first embodiment, compared with 0.7 pF in the first comparative example. The electrostatic capacitance C_c was 1.4 pF in the example of the first embodiment, compared with 1.0 pF in the first comparative example. The electrostatic capacitance C_d was 2.4 pF in the example of the first embodiment, compared with 3.9 pF in the first comparative example. The total of the electrostatic capacitances C_b to C_d was 4.6 pF in the example of the first embodiment, compared with 5.6 pF in the first comparative example.

(39) That is, the (total) electrostatic capacitance of the drive electrode **11a** and electrodes formed in the detection electrode layer **21** according to the example of the first embodiment is smaller than that according to the first comparative example. Moreover, the electrostatic capacitances C_b and C_c in the example of the first embodiment were respectively larger than the electrostatic capacitances C_b and C_c in the first comparative example.

(40) FIG. **8** is a graph showing a result of comparison between the intensity of a signal from a pressure detecting electrode **21a** according to the example of the first embodiment and the intensity of a signal from a pressure detecting electrode according to the first comparative example. A load was applied to each of the cover member **5** according to the example of the first embodiment and a cover member of the first comparative example, and the intensity of the signal from the pressure detecting electrode was measured. The measurement result shows that, when the load is 100 g or more and 700 g or less, the intensity of the signal according to the example of the first embodiment was larger than that according to the first comparative example. Hence, the measurement result shows that sensitivity in detecting pressure according to the example of the first embodiment is higher than that according to the first comparative example.

Second Embodiment

(41) Next, with reference to FIGS. **9** and **10**, a configuration of a display device **300** according to a second embodiment will be described. In the second embodiment, a floating electrode **321d** is disposed between a position detecting electrode **21b** and a reference potential electrode **321c**. FIG. **9** is a cross-sectional view of a touch panel **301** included in the display device **300** according to the

second embodiment. FIG. 10 is a plan view of a portion of a detection electrode layer 321 according to the second embodiment. Note that the same components as those of the first embodiment are denoted by the same reference signs as those of the first embodiment, and descriptions thereof are omitted.

(42) As illustrated in FIG. 9, the display device 300 includes the touch panel 301. The touch panel 301 includes the reference potential electrode 321c and the floating electrode 321d. The reference potential electrode 321c is included in the detection electrode layer 321 and disposed between the floating electrode 321d and the pressure detecting electrode 21a. The reference potential electrode 321c is connected to the ground, and has the potential of the ground. At least a portion of the floating electrode 321d is positioned to overlap with a drive electrode 11a in plan view. In addition, at least a portion of the floating electrode 321d is positioned not to overlap with a drive floating electrode 11b in plan view. The floating electrode 321d and the position detecting electrode 21b form an electrostatic capacitance C_f of a fringe component.

(43) As illustrated in FIG. 10, each of a plurality of the floating electrodes 321d is shaped into a rectangle in plan view. Moreover, the plurality of floating electrodes 321d are arranged side by side in the Y-direction. The plurality of floating electrodes 321d are not connected to each other. A width W7 of each floating electrode 321d is smaller than, for example, a width W5 of each position detecting electrode 21b. Furthermore, the width W7 is, for example, smaller than a width W31; that is, the largest width of the pressure detecting electrode 21a, and larger than a width W32; that is, the smallest width of the pressure detecting electrode 21a.

(44) According to the configuration of the second embodiment, the reference potential electrode 321c is disposed toward the pressure detecting electrode 21a with respect to the floating electrode 321d. Such a feature makes it possible to keep the reference potential electrode 321c from decreasing the intensity of a signal from the position detecting electrode 21b. As a result, the feature can reduce a difference between the intensity of the signal from the pressure detecting electrode 21a and the intensity of the signal from the position detecting electrode 21b. As a result, the intensity of the signal from the pressure detecting electrode 21a and the intensity of the signal from the position detecting electrode 21b can be easily adjusted on the touch panel controller. Note that other configurations and the advantageous effects are the same between the second embodiment and the first embodiment.

Result of Comparison Between Example of Second Embodiment and First Comparative Example

(45) Next, with reference to FIG. 11, a result of comparison is described between the display device 300 according to an example of the second embodiment and the display device 200 according to the first comparative example.

(46) The example of the second embodiment is the display device 300 illustrated in FIGS. 9 and 10. FIG. 11 is a table showing the result of comparison between the display device 300 according to the example of the second embodiment and the display device 200 according to the first comparative example. As to each of the display device 300 according to the example of the second embodiment and the display device 200 according to the first comparative example, a value of the electrostatic capacitance C_b between the drive electrode and the position detecting electrode, a value of the electrostatic capacitance C_c between the drive electrode and the pressure detecting electrode, and a value of the electrostatic capacitance C_d between the drive electrode and the reference potential electrode were measured. The electrostatic capacitance C_b was 1.1 pF in the example of the second embodiment, compared with 0.7 pF in the first comparative example. The electrostatic capacitance C_c was 1.1 pF in the example of the second embodiment, compared with 1.0 pF in the first comparative example. The electrostatic capacitance C_d was 2.7 pF in the example of the second embodiment, compared with 3.9 pF in the first comparative example. The total of the electrostatic capacitances C_b to C_d was 4.9 pF in the example of the second embodiment, compared with 5.6 pF in the first comparative example.

(47) That is, the (total) electrostatic capacitance of the drive electrode 11a and electrodes formed in

the detection electrode layer **321** according to the example of the second embodiment is smaller than that according to the first comparative example. Moreover, in the example of the second embodiment, the difference between the electrostatic capacitances C_b and C_c is 0 pF, which is lower than 0.3 pF in the first comparative example. That is, in the example of the second embodiment, a difference between the intensity of the signal from the pressure detecting electrode **21a** and the intensity of the signal from the position detecting electrode **21b** is reduced.

Third Embodiment

(48) Next, with reference to FIG. **12**, a configuration of a display device **400** according to a third embodiment will be described. In the third embodiment, a floating electrode **421d** is surrounded with a pressure detecting electrode **421a** in plan view. FIG. **12** is a plan view of a portion of the display device **400** according to the third embodiment. Note that the same components as those of the first or second embodiment are denoted by the same reference signs as those of the first or second embodiment, and descriptions thereof are omitted.

(49) As illustrated in FIG. **12**, the display device **400** includes a touch panel **401**. The touch panel **401** includes: a pressure detecting electrode **421a**; a position detecting electrode **421b**; a reference potential electrode **421c**; and a floating electrode **421d**. The pressure detecting electrode **421a** includes a plurality of pressure detecting units **461**. Each of the plurality of pressure detecting units **461** is shaped into a rectangle in plan view. The plurality of pressure detecting units **461** are connected to each other through a connection portion **461a**. A width $W2a$ of the connection portion **461a** in the X-direction is smaller than a width $W1a$ of the pressure detecting electrode **421a** in the X-direction.

(50) The position detecting electrode **421b** has a constant width in the X-direction; namely, a width $W3a$. The position detecting electrode **421b** extends in the Y-direction. Moreover, the width $W3a$ is smaller than the width $W1a$ of the pressure detecting electrode **421a** in the X-direction.

Furthermore, the reference potential electrode **421c** is formed between the pressure detecting electrode **421a** and the position detecting electrode **421b** in the X-direction. In addition, the reference potential electrode **421c** is disposed between the plurality of pressure detecting units **461** in the Y direction. A width $W4a$ of the reference potential electrode **421c** in the X-direction is smaller than the width $W1a$ of the pressure detecting electrode **421a**. The width $W4a$ is smaller than the width $W3a$ of the position detecting electrode **421b**.

(51) Moreover, the pressure detecting units **461** have holes formed in a matrix in plan view. Each of a plurality of the floating electrodes **421d** is disposed in a corresponding one of the plurality of holes formed in the pressure detecting units **461**. Thus, the plurality of floating electrodes **421d** are arranged in a matrix in plan view, and surrounded with the pressure detecting electrode **421a**. For example, 16 (four rows by four columns) floating electrodes **421d** are arranged in one pressure detecting unit **461**. Each of the floating electrodes **421d** has a width $W5a$ in the X-direction. The width $W5a$ is larger than the width $W4a$ and the width $W2a$. Furthermore, the width $W5a$ is smaller than the width $W3a$.

(52) Thanks to the configuration of the third embodiment, the floating electrodes **421d** can reduce the electrostatic capacitance between the drive electrodes and the electrodes formed in the detection electrode layer. Moreover, the floating electrodes **421d** are formed in the pressure detecting electrode **421a**. Such a feature makes it possible to reduce the intensity of a signal from a pressure detecting electrode, and to reduce a difference between the intensity of the signal from the pressure detecting electrode and the intensity of the signal from a position detecting electrode. Note that other configurations and the advantageous effects are the same between the third embodiment and the first and second embodiments.

Result of Comparison Between Example of Third Embodiment and Second Comparative Example

(53) Next, with reference to FIGS. **13** and **14**, a result of comparison is described between the display device **400** according to an example of the third embodiment and a display device **500** according to a second comparative example. Note that the display device **500** according to the

second comparative example is configured to confirm advantageous effects of the display device **400** according to the example of the third embodiment. The display device **500** does not represent the prior art. FIG. **13** is a view of a configuration of the display device **500** according to the second comparative example.

(54) The example of the third embodiment is the display device **400** illustrated in FIG. **12**.

Moreover, as illustrated in FIG. **13**, the display device **500** includes: a pressure detecting electrode **521a**; a position detecting electrode **521b**; and a reference potential electrode **521c**. In the second comparative example, the floating electrodes **421d** are omitted, and an area corresponding to the floating electrodes **421d** is provided with the pressure detecting electrode **521a**. The position detecting electrode **521b** and the reference potential electrode **521c** are configured respectively in the same manner as the position detecting electrode **421b** and the reference potential electrode **421c**.

(55) FIG. **14** is a table showing the result of comparison between the display device **500** according to the example of the third embodiment and the display device **400** according to the second comparative example. As to each of the display device **500** according to the example of the third embodiment and the display device **400** according to the second comparative example, a value of the electrostatic capacitance C_b between a drive electrode and the position detecting electrode, a value of the electrostatic capacitance C_c between the drive electrode and the pressure detecting electrode, and a value of the electrostatic capacitance C_d between the drive electrode and the reference potential electrode were measured. The electrostatic capacitance C_b was 0.8 pF in the example of the third embodiment, compared with 0.8 pF in the second comparative example. The electrostatic capacitance C_c was 2.4 pF in the example of the third embodiment, compared with 4.6 pF in the second comparative example. The electrostatic capacitance C_d was 0.2 pF in the example of the third embodiment, compared with 0.2 pF in the second comparative example. The total of the electrostatic capacitances C_b to C_d was 3.4 pF in the example of the third embodiment, compared with 5.6 pF in the second comparative example.

(56) That is, the (total) electrostatic capacitance of the drive electrode **11a** and electrodes formed in a detection electrode layer according to the example of the third embodiment is smaller than that according to the second comparative example. Moreover, in the example of the third embodiment, the difference between the electrostatic capacitances C_b and C_c is 1.6 pF, which is lower than 3.8 pF in the first comparative example. That is, in the example of the third embodiment, a difference between the intensity of a signal from the pressure detecting electrode **421a** and the intensity of a signal from the position detecting electrode **421b** is reduced.

Modifications

(57) The embodiments described above are mere examples for implementing the present disclosure. Therefore, the present disclosure shall not be limited to the above embodiments, and the above embodiments can be appropriately modified and implemented unless otherwise departing from the scope of the present disclosure. (1) The first to third embodiments show an example in which the floating electrode and the drive electrode are positioned to overlap with each other in plan view. However, the present disclosure shall not be limited to such an example. That is, only a portion of the floating electrode may be positioned to overlap with the drive electrode in plan view. Alternatively, the floating electrode may be positioned not to overlap with the drive electrode in plan view. (2) The first to third embodiments show an example in which the pressure detecting unit is shaped into a rectangle or a hexagon, the position detecting unit is shaped into a rectangle, and the floating electrode is shaped into a rectangle or a trapezoid. However, the present disclosure shall not be limited to such an example. For example, any of the pressure detecting unit, the position detecting unit, and the floating electrode may be shaped into a circle or a polygon. (3) The first to third embodiments show an example in which the potential of the reference potential electrode is the potential of the ground. However, the present disclosure shall not be limited to such an example. For example, the potential of the reference potential electrode may be set to a

predetermined constant potential different from the potential of the ground. (4) The first to third embodiments show an example in which the direction in which the plurality of drive electrodes are arranged (the Y-direction) is orthogonal to the direction in which the plurality of pressure detecting electrodes are arranged (the X-direction). However, the present disclosure shall not be limited to such an example. For example, the direction in which the plurality of drive electrodes are arranged may intersect with the direction in which the plurality of pressure detecting electrodes are arranged at an angle of less than 90°. (5) The first to third embodiments show an example in which the drive electrodes, the drive floating electrodes, the pressure detecting electrodes, the position detecting electrodes, and the floating electrodes are made of ITO. However, the present disclosure shall not be limited to such an example. For example, the drive electrodes, the drive floating electrodes, the pressure detecting electrodes, the position detecting electrodes, and the floating electrodes may be made of a conductive metal material. (6) The first to third embodiments show an example in which the touch panel is provided with the drive floating electrodes. However, the present disclosure shall not be limited to such an example. That is, the touch panel may omit the drive floating electrodes. (7) The first to third embodiments show an example in which the touch panel is provided to the display device. However, the present disclosure shall not be limited to such an example. That is, the touch panel of the present disclosure may be provided to a device having no display unit. (58) The above capacitive touch panel and display device can also be described as follows. (59) A capacitive touch panel according to a first configuration detects: a position included in a touch surface and touched with a pointing object; and pressure applied on the touch surface with the pointing object. The capacitive touch panel includes: a drive electrode supplied with a drive signal; a detection electrode disposed across the drive electrode from the touch surface, and including a position detecting electrode that detects the position included in the touch surface and touched with the pointing object, and a pressure detecting electrode that detects the pressure applied on the touch surface with the pointing object; and a floating electrode having a floating potential. The position detecting electrode, the pressure detecting electrode, and the floating electrode are formed in a layer (a first configuration). (60) According to the first configuration, the floating electrode having a floating potential is formed in the same layer as the layer in which a detection electrode is formed. Between the drive electrode and the floating electrode, an electrostatic capacity is less likely to be formed. Hence, even if the capacitive touch panel has a pressure detecting function, such a feature makes it possible to prevent the electrostatic capacity from increasing between the drive electrode and an electrode that is formed in the layer in which a detection electrode is formed. (61) In the first configuration, at least a portion of the floating electrode may be positioned to overlap with the drive electrode in plan view (a second configuration). (62) Here, the drive electrode and an electrode, which is included in the detection electrode layer and positioned to overlap with the drive electrode in plan view, form a parallel plate capacitor. On the other hand, the drive electrode and an electrode, which is included in the detection electrode layer and positioned at least partially not to overlap with the drive electrode, form an electrostatic capacitance (an electrostatic capacitance of a fringe component) between an end portion of the drive electrode and an end portion of the electrode included in the detection electrode layer. An electrostatic capacitance formed by the parallel plate capacitor is larger than the electrostatic capacitance of the fringe component. Thanks to the second configuration, at least a portion of an electrode, which is included in the detection electrode layer and positioned to overlap with the drive electrode in plan view, acts as the floating electrode. Such a feature can reduce the size of the parallel plate capacitor. As a result, the feature can prevent an increase in the electrostatic capacitance between the drive electrode and the electrode that is formed in the detection electrode layer. (63) In the first or second configuration, the capacitive touch panel may further include a drive floating electrode formed in a same layer as the drive electrode is formed, and having the floating

potential. The position detecting electrode may be positioned to overlap with the drive floating electrode in plan view, and the floating electrode may be positioned not to overlap with the drive floating electrode in plan view (a third configuration).

(64) Here, at least a portion of the position detecting electrode is positioned not overlap with the drive electrode in plan view, in order to be capacitively coupled to the pointing object. Hence, between an end portion of the drive electrode and the position detecting electrode, an electrostatic capacitance due to a fringe component is formed. However, an electrostatic capacitance of the parallel plate capacitor formed of the pressure detecting electrode and the drive electrode is larger than the electrostatic capacitance, due to the fringe component, formed between the position detecting electrode and the drive electrode. Hence, the intensity of a signal from the pressure detecting electrode is larger than the intensity of a signal from the position detecting electrode, so that the adjustment is complex between the intensity of the signal from the pressure detecting electrode and the intensity of the signal from the position detecting electrode. Thanks to the third configuration, the drive electrode and the drive floating electrode are capacitively coupled together, and the drive floating electrode and the position detecting electrode are capacitively coupled together. Thanks to such a feature, even if the drive electrode and the position detecting electrode are positioned not to overlap with each other in plan view, the drive electrode and the position detecting electrode can form an electrostatic capacitance through the drive floating electrode. As a result, the feature can increase the intensity of a signal for detecting the touch position based on the electrostatic capacitance. Moreover, the floating electrode is positioned not to overlap with the drive floating electrode in plan view. Thus, the position detecting electrode, which is positioned to overlap with the drive floating electrode in plan view, can be formed large in size. These features make it possible to reduce a difference between the intensity of a signal from the pressure detecting electrode and the intensity of a signal from the position detecting electrode, so that the adjustment can be simplified between the intensity of the signal from the pressure detecting electrode and the intensity of the signal from the position detecting electrode.

(65) In the first or second configuration, the capacitive touch panel may further include: a drive floating electrode formed in a same layer as the drive electrode is formed, and having the floating potential; and a reference potential electrode connected to a reference potential (a fourth configuration).

(66) Here, if the drive floating electrode and the pressure detecting electrode are capacitively coupled to each other, the capacitance inevitably changes just because when the pointing object touches the touch panel even if the capacitive touch panel is not pressed by the pointing object. In this case, it is difficult to distinguish the capacitance change caused merely by a touch of the pointing object from the capacitance change caused by a press of the pointing object. On the other hand, according to the fourth configuration, the reference potential electrode can prevent capacitive coupling between the drive floating electrode and the pressure detecting electrode. Hence, the drive electrode and the pressure detecting electrode exhibit the capacitance change only when the touch panel is pressed. As a result, the fourth configuration can accurately distinguish the capacitance change caused merely by a touch of the pointing object from the capacitance change caused by a press of the pointing object.

(67) In any one of the first to fourth configurations, the capacitive touch panel may further include a reference potential electrode connected to a reference potential. In the layer in which the floating electrode is formed, the reference potential electrode may be disposed between the floating electrode and the position detecting electrode (a fifth configuration).

(68) Thanks to the fifth configuration, the detection electrode layer is provided with the floating electrode. Such a feature makes it possible to prevent an increase in the electrostatic capacitance formed by the reference potential electrode and the drive electrode.

(69) In any one of the first to fourth configurations, the capacitive touch panel may further include a reference potential electrode connected to a reference potential. In the layer in which the floating

electrode is formed, the reference potential electrode may be disposed between the floating electrode and the pressure detecting electrode (a sixth configuration).

(70) Thanks to the sixth configuration, the reference potential electrode is disposed toward the pressure detecting electrode with respect to the floating electrode. Such a feature makes it possible to reduce a decrease in the signal intensity from the position detecting electrode caused by the reference potential electrode. As a result, the feature can reduce a difference between the intensity of the signal from the pressure detecting electrode and the intensity of the signal from the position detecting electrode.

(71) In any one of the first to fourth configurations, the floating electrode may be surrounded with the pressure detecting electrode in plan view (a seventh configuration).

(72) Thanks to the seventh configuration, the floating electrode can reduce an electrostatic capacitance between the drive electrode and an electrode formed in the detection electrode layer. Moreover, the floating electrode is formed in the pressure detecting electrode. Such a feature makes it possible to reduce the intensity of a signal from the pressure detecting electrode, and to reduce a difference between the intensity of the signal from the pressure detecting electrode and the intensity of the signal from the position detecting electrode.

(73) A display device according to an eighth configuration includes a capacitive touch panel that detects: a position included in a touch surface and touched with a pointing object; and pressure applied on the touch surface with the pointing object; and a display that displays an image. The capacitive touch panel includes: a drive electrode supplied with a drive signal; a detection electrode disposed across the drive electrode from the touch surface, and including a position detecting electrode that detects the position included in the touch surface and touched with the pointing object, and a pressure detecting electrode that detects the pressure applied on the touch surface with the pointing object; and a floating electrode having a floating potential. The position detecting electrode, the pressure detecting electrode, and the floating electrode are formed in a layer (an eighth configuration).

(74) The eighth configuration makes it possible to provide a display device having a pressure detecting function capable of preventing an increase in electrostatic capacitance between the drive electrode and an electrode that is formed in the layer in which a detection electrode is formed.

Claims

1. A capacitive touch panel that detects a position included in a touch surface and touched with a pointing object, and a pressure applied on the touch surface with the pointing object, the capacitive touch panel comprising: a drive electrode supplied with a drive signal; a detection electrode disposed across the drive electrode from the touch surface, and including a position detecting electrode configured to detect the position included in the touch surface and touched with the pointing object, and a pressure detecting electrode configured to detect the pressure applied on the touch surface with the pointing object; a floating electrode having a floating potential, the floating electrode not connected to another electrode, not connected to a voltage source, and not connected to ground; and a reference potential electrode having a reference potential, wherein the reference potential is equal to a potential of the ground, wherein the position detecting electrode, the pressure detecting electrode, and the floating electrode are formed in a layer, and in the layer, the position detecting electrode, the reference potential electrode, the floating electrode, and the pressure detecting electrode are disposed adjacent to each other in the mentioned order.
2. The capacitive touch panel according to claim 1, wherein at least a portion of the floating electrode is positioned to overlap with the drive electrode in a plan view.
3. The capacitive touch panel according to claim 2, further comprising: a drive floating electrode formed in a same layer as the drive electrode is formed, and having the floating potential, wherein the position detecting electrode is positioned to overlap with the drive floating electrode in the plan

view, and the floating electrode is positioned not to overlap with the drive floating electrode in the plan view.

4. The capacitive touch panel according to claim 1, further comprising: a drive floating electrode formed in a same layer as the drive electrode is formed, and having the floating potential.

5. The capacitive touch panel according to claim 1, wherein the floating electrode is surrounded with the pressure detecting electrode in a plan view.

6. A capacitive touch panel that detects a position included in a touch surface and touched with a pointing object, and a pressure applied on the touch surface with the pointing object, the capacitive touch panel comprising: a drive electrode supplied with a drive signal; a detection electrode disposed across the drive electrode from the touch surface, and including a position detecting electrode configured to detect the position included in the touch surface and touched with the pointing object, and a pressure detecting electrode configured to detect the pressure applied on the touch surface with the pointing object; a floating electrode having a floating potential, the floating electrode not connected to another electrode, not connected to a voltage source, and not connected to ground; and a reference potential electrode having a reference potential, wherein the reference potential is equal to a potential of the ground, wherein the position detecting electrode, the pressure detecting electrode, and the floating electrode are formed in a layer, and in the layer, the position detecting electrode, the floating electrode, the reference potential electrode, and the pressure detecting electrode are disposed adjacent to each other in the mentioned order.

7. A display device, comprising: a capacitive touch panel configured to detect a position included in a touch surface and touched with a pointing object, and a pressure applied on the touch surface with the pointing object; and a display configured to display an image, wherein the capacitive touch panel includes: a drive electrode supplied with a drive signal; a detection electrode disposed across the drive electrode from the touch surface, and including a position detecting electrode configured to detect the position included in the touch surface and touched with the pointing object, and a pressure detecting electrode configured to detect the pressure applied on the touch surface with the pointing object; a floating electrode having a floating potential, the floating electrode not connected to another electrode, not connected to a voltage source, and not connected to ground; and a reference potential electrode having a reference potential, wherein the reference potential is equal to a potential of the ground, the position detecting electrode, the pressure detecting electrode, and the floating electrode are formed in a layer, and in the layer, the position detecting electrode, the reference potential electrode the floating electrode, and the pressure detecting electrode are disposed adjacent to each other in the mentioned order.
